Hotelling's tube test

and its generalization to test when a nuisance parameter is only present under the alternative

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Model:

$Y_i = \beta \frac{X_i^\lambda - 1}{\lambda} + u_i$ where $u_i \sim N(0, \sigma^2)$.

Test:

$H_0 : \beta(\lambda) = 0$ vs. $H_a : \beta(\lambda) \neq 0$

Feature:

(1) λ is nuisance parameter not present under the null. (2) If λ were known, then LRT, Wald and Score test could be used and we know their limiting distribution.

Idea:

$\underset{\lambda}{sup} LRT(\lambda)$, $\underset{\lambda}{sup} T_n(\lambda)$

Issues:

Find critical value $\Rightarrow$ Hotelling's tube formula

LRT

$$
\begin{aligned}
SLRT = \sup_{\lambda} LRT(\lambda) &= 2\sup_{\lambda}(l(\lambda,\hat{\beta}) - l(\lambda,0)) \\
&= -\inf_{\lambda} \sum_i \frac{(y_i - h_i(\lambda)\hat{\beta}(\lambda))^2}{\sigma^2} + \sum_i \frac{y_i^2}{\sigma^2} \\
&= -\sum \frac{y_i^2}{\sigma^2}[\inf_{\lambda} \frac{\sum(y_i - h_i(\lambda)\hat{\beta}(\lambda))^2}{\sum y_i^2} - 1] \\
&= -\sum \frac{y_i^2}{\sigma^2}[1 - \sup_{\lambda}(\gamma(\lambda)^T U)^2 - 1] \\
&= \sum \frac{y_i^2}{\sigma^2} \sup_{\lambda}(\gamma(\lambda)^T U)^2
\end{aligned}
$$

Observing $R^2 = \sum_i \frac{y_i^2}{\sigma^2} \sim \chi_n^2$. we have $\sqrt{SLRT} = RW$ and $R \perp W$. Knowles (1987)'s upper bound can be used.

$$
P(\sqrt{SLRT} > c) \leq \frac{|\gamma|}{2\pi} e^{-c^2/2} + 1 - \Phi(c)
$$

Score Test

Score function for β under the null model

$$\frac{\partial l}{\partial \beta}|_{\beta=0} = \sum_i \frac{(y_i - h_i(\lambda)\beta)h_i(\lambda)}{\sigma^2}|_{\beta=0}$$
$$= \sum_i \frac{y_i h_i(\lambda)}{\sigma^2}$$

The normalized score test statistics is

$$T_n = \frac{\sum y_i h_i(\lambda)/\sigma^2}{\sqrt{\sum h_i^2(\lambda)/\sigma^2}}$$
$$= \frac{\frac{1}{\sigma} y^T h(\lambda)}{\|h(\lambda)\|}$$

Observing $T = \underset{\lambda}{sup} T_n(\lambda) = RW$, recall $W = \underset{\lambda}{sup}(\frac{h(\lambda)}{\|h(\lambda)\|})^T \frac{Y}{\|Y\|}$.

$P(|T| > c) \leq 2P(T > c) \leq \frac{|\gamma|}{\pi} e^{-c^2/2} + 2(1 - \Phi(c))$

Simulation: $\lambda \in (0.5, 0, -0.5)$, $x \sim Unif[0, 10]$, $\beta \in (0, 0.5, 1, 3)$ rescaled by $\sqrt{n}$.

Table: Tube Test Performance

$n =$	50	100	200	300	500	1000
$\lambda = 0.5$						
$\beta = 0$	0.056	0.046	0.064	0.052	0.042	0.064
$\beta = 0.5$	0.182	0.184	0.166	0.150	0.136	0.172
$\beta = 1$	0.402	0.380	0.396	0.376	0.360	0.368
$\beta = 3$	0.988	0.984	1.000	0.990	0.986	1.000
$\lambda = 0$						
$\beta = 0$	0.054	0.044	0.052	0.040	0.040	0.046
$\beta = 0.5$	0.150	0.100	0.118	0.100	0.090	0.122
$\beta = 1$	0.232	0.266	0.234	0.186	0.230	0.200
$\beta = 3$	0.848	0.850	0.870	0.872	0.852	0.864
$\lambda = -0.5$						
$\beta = 0$	0.056	0.036	0.058	0.060	0.052	0.056
$\beta = 0.5$	0.162	0.150	0.178	0.166	0.178	0.166
$\beta = 1$	0.276	0.308	0.374	0.362	0.360	0.402
$\beta = 3$	0.768	0.800	0.876	0.910	0.934	0.954

A mixture model: $f(x_i; 0, \theta) = (1-r)\phi(x_i; 0) + r\phi(x_i; \theta)$

Test: $H_0 : r = 0$ vs $H_a : r \neq 0$.

LRT: $\sup_\theta LRT(\theta) = 2\sum_i log[1 + \sup_\theta r(\theta)(e^{x_i\theta - \frac{\theta^2}{2}} - 1)]$

Score: $\sup_\theta T_n(\theta) = \sup_\theta \frac{\sum_i (e^{x_i\theta - \frac{\theta^2}{2}} - 1)}{\sqrt{n(e^{\theta^2} - 1)}}$

Guess Individual normalized score $\frac{e^{x_i\theta - \theta^2/2} - 1}{\sqrt{var(e^{x_i\theta - \theta^2/2} - 1)}}$ could be thought of as a curve on S^{n-1}.

simulation $\theta \in (0, 1, 3, 5)$ rescaled by $n^{1/4}$, $r = 0.3$

$n =$	50	100	200	300	500	1000
$\theta = 0$	0.020	0.030	0.016	0.014	0.024	0.014
$\theta = 1$	0.090	0.052	0.044	0.048	0.040	0.060
$\theta = 3$	0.574	0.624	0.710	0.752	0.872	0.968
$\theta = 5$	0.988	1.000	1.000	1.000	1.000	1.000